

Nandan Kamath

nandank@umich.edu | (248) 974-1327 | [LinkedIn](#) | [GitHub](#) | [Personal Portfolio](#)

EDUCATION

University of Michigan, Ann Arbor, MI

August 2023 - May 2027

BSE, Computer Engineering

- Coursework: Very Large Scale Integrated Design (VLSI), Parallel Computer Architecture, Computer Architecture, Microarchitecture, Digital Integrated Circuits, Networks, Computer Organization, Logic Design, Signals and Systems, Data Structures and Algorithms

SKILLS

- Computer Programming: C/C++, SystemVerilog, Python, TCL, Linux
- Engineering Tools: Synopsys (Verdi, Design Compiler), Cadence (Virtuoso, JasperGold, Innovus), ModelSim, HSPICE
- VLSI/Architecture: CPU & GPU Architecture, OoO Pipelines, SoC Design, Memory Hierarchy, Functional DV, PPA Optimization, Cache Coherency, Synthesis & P&R, Static Timing Analysis, DRC/LVS, UVM, ARM/RISC-V ISA, RTL, Clock Domain Crossing

TECHNICAL PROJECTS

R10K Style Out-of-Order Processor

August 2025 - December 2025

- Architected and verified an 83.33 MHz N-way superscalar out-of-order MIPS R10K CPU in SystemVerilog using Synopsys Design Compiler and Verdi, implementing dynamic scheduling, register renaming via N-ported Map Table, and 2N-depth Instruction Queue
- Engineered the backend execution engine to support N-way dispatch and in-order retirement by integrating a Reservation Station and Reorder Buffer (ROB), achieving speculative state recovery via a GShare branch predictor utilizing a Pattern History Table and BTB.
- Minimized memory latency by designing a non-blocking instruction cache with stream-buffer prefetching and a 4-way set-associative data cache, utilizing Miss Status Handling Registers (MSHRs) and a sequentially ordered Store Queue to enforce memory ordering.

VLSI Design & Optimization of a 16-Bit RISC Processor

January 2026 - April 2026

- Engineered the physical layout of a 16-bit RISC processor in Cadence Virtuoso and Innovus, designing custom schematics and optimized layouts for the ALU, Logarithmic Shifter, Register File, Decoder, and Program Counter that passed all DRC/LVS requirements
- Optimized ALU critical-path performance by implementing a 16-bit Kogge-Stone adder, leveraging $O(\log_2 N)$ carry logic and reusing generate/propagate hardware to decrease latency by 23% (avg. 272 ps improvement in HSPICE) and a 32% reduction in silicon area.
- Validated system functionality through a modular testing strategy using NC-Verilog, performing mixed-signal verification to ensure the processor's logic between the behavioral model and the extracted Cadence transistor netlist was equivalent across all instructions

Hardware Acceleration for Real-time Semantic Graph SLAM

February 2026 - April 2026

- Created custom acceleration kernels through the AIE-ML library for an AMD Versal SoC pivoting from a space constrained FPGA fabric to accelerate RangeNet++ SLAM by mapping convolutions to tiled INT8 matrix multiplications processed as 32-channel blocks
- Profiled and accelerated the compute-intensive convolutional layers of the RangeNet++ architecture, compressing the memory footprint by 4x (201 MB to 50 MB) via batch-norm folding and per-channel scaling to enable real-time 3D LiDAR semantic segmentation.
- Checked cycle-approximate AIE simulations against an NVIDIA A10 GPU baseline, demonstrating a 14.5x speedup (231.4 ms to 16.0 ms) and a 120.6x energy efficiency gain (222.0 W to 12.7 W) meeting the goal of a strict 10 Hz LiDAR real-time constraints.

CMOS Adder Design

October 2025 - November 2025

- Created two full-custom 8-bit dual-mode ripple-carry adder schematics in Cadence Virtuoso using complementary pass transistor logic for high clock speed and static CMOS for low power, achieving a 2 GHz clock speed and 151 μ W performance targets, respectively.
- Optimized device sizing, critical-path, and node capacitance across $> 180 \mu\text{m}$ of transistor width, integrating transmission gate registers and pass-transistor multiplexers to meet strict timing and functional correctness in both adder and accumulation operations.
- Verified functionality and timing through Cadence transient simulations, as well as random and custom-designed test benches, validating correct circuit operation across all mode transitions and measuring power and delay tradeoffs between the two logic families.

Sequential Four-Function Calculator

December 2024 - January 2025

- Implemented a sequential four-function calculator in Verilog on the Altera DE2-115 FPGA board using Quartus, enabling addition, subtraction, multiplication, and division of two's complement 11-bit integers with real-time user input via sliders and push buttons
- Developed a custom finite state machine controller and integrated an arithmetic datapath incorporating an adder, subtractor, Booth's algorithm multiplier, and repeated subtraction division modules, with accurate operation, overflow detection, and display of result
- Simulated and debugged calculator operations in ModelSim's waveform viewer by verifying all controller state transitions and datapath relations, and ensured stable 50MHz clock operation by resolving timing issues and handling edge cases like overflow, divide-by-zero

PROFESSIONAL EXPERIENCE

Walmart Inc.

Bentonville, AR

Automation Engineering Intern

June 2025 - August 2025

- Conducted data analysis using SQL and Google Cloud Platform (GCP) to identify throughput constraints in a fulfillment center, revealing that optimizing the tote consolidation process would add 100,000 cubic feet of storage, equivalent to a 2% gain in volume
- Built an Integer Linear Programming (ILP) optimization tool in Python to assign associates to workstations based on volume targets at a given time, availability, and productivity, minimizing headcount while consistently achieving 94–106% of outbound demand
- Oversaw installation and commissioning of next-generation robotic consolidation system across multiple sites, used data analysis and projections to estimate current throughput required, and determined that 22 robots would be required to meet demand through FY31